

Supplementary Information

Tables S1–S17 and Figures S1–S11

Dynamic Evaluation of Governance Strategies for Mitigating Social Media Data Pollution: An Agent-Based Modeling Approach

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1 Supplementary tables

Reading the tables

Tables S1–S8 report condition summaries; Tables S9–S17 report prespecified comparisons. Estimates and their 95% intervals appear together, except in Table S10, which uses separate interval columns. Each condition uses 30 independent runs; repeated appearances of an identical condition reuse the same runs.

Condition summaries use Student- t confidence intervals. Comparisons use marginal, unadjusted Welch intervals, except for Table S10: its dimensionless PAWN-index differences have separate row-bootstrap stability and within-point Monte Carlo intervals, based on 500 LHS points per regime and initial pollution level, with 30 runs per point. These two interval types are not combined.

Outcome levels are percentages and their differences are percentage points unless stated otherwise. Network diagnostics and review burdens are counts; review totals include repeated reviews. Table notes specify subtraction order, units, valid sample sizes and exceptions. Unrounded values and identifiers are retained in the accompanying CSV files.

Table S1 Realized network diagnostics

ID	p_0 (%)	Regime	Network	N	Mean degree	Maximum degree	Isolates
1	10	No governance	BA	1000	3.99 [3.99, 3.99]	84.13 [77.50, 90.77]	0.00 [0.00, 0.00]
70	10	No governance	ER	1000	3.98 [3.95, 4.01]	11.97 [11.58, 12.35]	18.13 [16.61, 19.65]
71	10	No governance	WS	1000	4.00 [4.00, 4.00]	6.63 [6.40, 6.86]	0.00 [0.00, 0.00]
8	10	Legal	BA	1000	3.99 [3.99, 3.99]	80.10 [73.28, 86.92]	0.00 [0.00, 0.00]
72	10	Legal	ER	1000	4.00 [3.96, 4.04]	11.73 [11.34, 12.12]	17.30 [15.86, 18.74]
73	10	Legal	WS	1000	4.00 [4.00, 4.00]	6.70 [6.50, 6.90]	0.00 [0.00, 0.00]
35	10	Platform	BA	1000	3.99 [3.99, 3.99]	84.03 [77.15, 90.91]	0.00 [0.00, 0.00]
74	10	Platform	ER	1000	4.01 [3.98, 4.04]	11.70 [11.28, 12.12]	17.87 [16.64, 19.09]
75	10	Platform	WS	1000	4.00 [4.00, 4.00]	6.67 [6.44, 6.89]	0.00 [0.00, 0.00]
2	30	No governance	BA	1000	3.99 [3.99, 3.99]	86.87 [80.68, 93.06]	0.00 [0.00, 0.00]
76	30	No governance	ER	1000	4.01 [3.96, 4.05]	11.80 [11.38, 12.22]	18.57 [16.69, 20.44]
77	30	No governance	WS	1000	4.00 [4.00, 4.00]	6.77 [6.51, 7.02]	0.00 [0.00, 0.00]
17	30	Legal	BA	1000	3.99 [3.99, 3.99]	83.57 [77.66, 89.47]	0.00 [0.00, 0.00]
78	30	Legal	ER	1000	3.99 [3.95, 4.03]	11.83 [11.49, 12.17]	18.50 [16.69, 20.31]
79	30	Legal	WS	1000	4.00 [4.00, 4.00]	6.70 [6.48, 6.92]	0.00 [0.00, 0.00]
44	30	Platform	BA	1000	3.99 [3.99, 3.99]	86.50 [79.78, 93.22]	0.00 [0.00, 0.00]
80	30	Platform	ER	1000	4.03 [4.00, 4.06]	11.97 [11.55, 12.39]	18.27 [16.29, 20.24]
81	30	Platform	WS	1000	4.00 [4.00, 4.00]	6.70 [6.48, 6.92]	0.03 [-0.03, 0.10]
3	50	No governance	BA	1000	3.99 [3.99, 3.99]	78.13 [73.34, 82.92]	0.00 [0.00, 0.00]
82	50	No governance	ER	1000	3.98 [3.94, 4.03]	11.73 [11.37, 12.10]	17.60 [16.32, 18.88]
83	50	No governance	WS	1000	4.00 [4.00, 4.00]	6.53 [6.34, 6.72]	0.00 [0.00, 0.00]
26	50	Legal	BA	1000	3.99 [3.99, 3.99]	79.13 [72.81, 85.46]	0.00 [0.00, 0.00]
84	50	Legal	ER	1000	3.98 [3.94, 4.01]	11.57 [11.17, 11.97]	18.23 [16.45, 20.02]
85	50	Legal	WS	1000	4.00 [4.00, 4.00]	6.60 [6.33, 6.87]	0.00 [0.00, 0.00]
53	50	Platform	BA	1000	3.99 [3.99, 3.99]	85.53 [79.98, 91.08]	0.00 [0.00, 0.00]
86	50	Platform	ER	1000	3.98 [3.95, 4.02]	11.90 [11.46, 12.34]	17.80 [15.76, 19.84]
87	50	Platform	WS	1000	4.00 [4.00, 4.00]	6.57 [6.33, 6.80]	0.00 [0.00, 0.00]
88	10	No governance	BA	5000	4.00 [4.00, 4.00]	187.70 [169.85, 205.55]	0.00 [0.00, 0.00]
89	10	Legal	BA	5000	4.00 [4.00, 4.00]	187.07 [172.98, 201.15]	0.00 [0.00, 0.00]
90	10	Platform	BA	5000	4.00 [4.00, 4.00]	188.97 [171.64, 206.30]	0.00 [0.00, 0.00]
91	30	No governance	BA	5000	4.00 [4.00, 4.00]	188.20 [173.55, 202.85]	0.00 [0.00, 0.00]
92	30	Legal	BA	5000	4.00 [4.00, 4.00]	174.70 [158.83, 190.57]	0.00 [0.00, 0.00]
93	30	Platform	BA	5000	4.00 [4.00, 4.00]	175.77 [163.88, 187.65]	0.00 [0.00, 0.00]
94	50	No governance	BA	5000	4.00 [4.00, 4.00]	182.83 [168.10, 197.57]	0.00 [0.00, 0.00]
95	50	Legal	BA	5000	4.00 [4.00, 4.00]	177.57 [161.82, 193.31]	0.00 [0.00, 0.00]
96	50	Platform	BA	5000	4.00 [4.00, 4.00]	173.97 [163.18, 184.75]	0.00 [0.00, 0.00]

Note. Means with pointwise 95% Student- t confidence intervals in brackets, from 30 independent network realizations per condition; intervals are not clipped. p_0 is a percentage and N is the number of nodes. Degree counts incident links; isolates are nodes of degree zero. These diagnostics are counts, not percentages. BA, ER and WS denote Barabási–Albert, Erdős–Rényi and Watts–Strogatz networks. Unrounded results, condition IDs and sample sizes are in `tableS_network.csv`. Additional diagnostics are in `network_summary.csv`.

Table S2 Outcomes under fixed and state-responsive platform moderation

ID	p_0 (%)	Workload	κ	$\bar{E}$	$O(50)$	$P_R(50)$	n_{PR}
35	10	Fixed	1	99.73 [99.71, 99.75]	99.74 [99.67, 99.82]	2.44 [2.32, 2.56]	30
97	10	Responsive	1	75.00 [74.75, 75.25]	39.83 [39.06, 40.60]	46.78 [46.51, 47.04]	30
98	10	Responsive	2	84.09 [83.81, 84.38]	49.92 [49.53, 50.31]	32.21 [31.93, 32.49]	30
99	10	Responsive	5	92.58 [92.47, 92.68]	59.75 [58.97, 60.53]	17.89 [17.65, 18.13]	30
44	30	Fixed	1	99.26 [99.23, 99.30]	99.65 [99.59, 99.72]	6.39 [6.21, 6.57]	30
100	30	Responsive	1	69.41 [69.19, 69.63]	47.79 [47.41, 48.16]	49.24 [48.98, 49.50]	30
101	30	Responsive	2	81.59 [81.39, 81.79]	54.68 [54.31, 55.06]	34.22 [33.98, 34.46]	30
102	30	Responsive	5	91.95 [91.87, 92.03]	65.63 [64.84, 66.43]	21.59 [21.30, 21.88]	30
53	50	Fixed	1	98.84 [98.81, 98.88]	99.68 [99.63, 99.74]	9.52 [9.37, 9.68]	30
103	50	Responsive	1	66.52 [66.27, 66.78]	50.19 [49.78, 50.60]	50.59 [50.35, 50.84]	30
104	50	Responsive	2	79.84 [79.69, 80.00]	55.79 [55.30, 56.28]	36.49 [36.30, 36.68]	30
105	50	Responsive	5	91.71 [91.64, 91.79]	66.81 [65.96, 67.66]	24.73 [24.47, 24.99]	30

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 and outcomes are percentages. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios; n_{PR} is their number (30 throughout). Fixed coverage is 75% per step, also the responsive ceiling; TPR = TNR = 80%. Both modes use degree targeting; κ is inactive in fixed mode. All networks are BA with $N = 1000$. Unrounded results, condition IDs and sample sizes are in `tableS1_P3.csv`.

Table S3 Review burden under fixed and state-responsive platform moderation

ID	p_0 (%)	Workload	κ	Reviews	Clean reviews
35	10	Fixed	1	37500.00 [37500.00, 37500.00]	37265.47 [37253.24, 37277.69]
97	10	Responsive	1	14606.73 [14448.33, 14765.14]	11965.30 [11855.07, 12075.53]
98	10	Responsive	2	19149.60 [18798.70, 19500.50]	17125.20 [16819.97, 17430.43]
99	10	Responsive	5	23389.83 [23043.83, 23735.83]	22180.33 [21858.47, 22502.19]
44	30	Fixed	1	37500.00 [37500.00, 37500.00]	36874.73 [36852.83, 36896.64]
100	30	Responsive	1	18117.03 [17985.13, 18248.94]	14585.43 [14484.38, 14686.48]
101	30	Responsive	2	22561.60 [22311.22, 22811.98]	19962.00 [19743.16, 20180.84]
102	30	Responsive	5	25271.33 [25054.72, 25487.95]	23638.60 [23432.19, 23845.01]
53	50	Fixed	1	37500.00 [37500.00, 37500.00]	36536.90 [36517.79, 36556.01]
103	50	Responsive	1	19995.53 [19844.34, 20146.73]	15923.03 [15801.17, 16044.89]
104	50	Responsive	2	24677.20 [24477.06, 24877.34]	21588.73 [21419.50, 21757.97]
105	50	Responsive	5	25530.17 [25304.19, 25756.15]	23599.20 [23395.49, 23802.91]

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. The Reviews column counts all review events over steps 1–50, including repeated reviews of an account. The Clean reviews column counts events involving accounts that were clean when reviewed (TN+FP). Both columns are event counts, not percentages or numbers of distinct accounts. p_0 is a percentage. Fixed coverage is 75% per step, also the responsive ceiling; TPR = TNR = 80%. Both modes use degree targeting; κ is inactive in fixed mode. All networks are BA with $N = 1000$. Unrounded results, condition IDs and sample sizes are in `tableS1_P3_workload.csv`.

Table S4 Legal outcomes under random and degree-targeted selection

ID	p_0 (%)	Selection	L (%)	$\bar{E}$	$O(50)$
109	10	Random	5	15.61 [15.39, 15.83]	0.46 [0.38, 0.53]
106	10	Targeted	5	15.75 [15.51, 16.00]	0.23 [0.18, 0.27]
110	10	Random	10	19.18 [18.87, 19.49]	0.99 [0.86, 1.12]
107	10	Targeted	10	18.98 [18.66, 19.31]	0.57 [0.49, 0.65]
111	10	Random	20	25.92 [25.72, 26.12]	2.04 [1.89, 2.18]
108	10	Targeted	20	26.28 [25.85, 26.70]	1.73 [1.59, 1.87]
58	10	Random	70	67.23 [66.79, 67.67]	15.44 [14.94, 15.94]
8	10	Targeted	70	77.86 [77.39, 78.34]	18.39 [17.98, 18.81]
121	30	Random	5	10.29 [10.17, 10.40]	0.27 [0.20, 0.34]
118	30	Targeted	5	9.36 [9.23, 9.49]	0.12 [0.08, 0.16]
122	30	Random	10	13.82 [13.66, 13.97]	0.63 [0.54, 0.72]
119	30	Targeted	10	11.89 [11.74, 12.04]	0.31 [0.25, 0.37]
123	30	Random	20	21.14 [20.94, 21.34]	1.64 [1.51, 1.78]
120	30	Targeted	20	18.32 [18.16, 18.49]	1.07 [0.98, 1.17]
60	30	Random	70	64.52 [64.24, 64.80]	13.64 [13.30, 13.98]
17	30	Targeted	70	73.59 [73.19, 74.00]	16.18 [15.83, 16.53]
133	50	Random	5	7.41 [7.30, 7.53]	0.19 [0.14, 0.24]
130	50	Targeted	5	5.97 [5.86, 6.08]	0.05 [0.02, 0.08]
134	50	Random	10	11.11 [11.01, 11.21]	0.39 [0.30, 0.48]
131	50	Targeted	10	8.09 [8.01, 8.18]	0.15 [0.11, 0.19]
135	50	Random	20	18.53 [18.37, 18.69]	1.04 [0.92, 1.16]
132	50	Targeted	20	14.05 [13.93, 14.17]	0.77 [0.67, 0.87]
62	50	Random	70	62.96 [62.71, 63.21]	12.13 [11.74, 12.53]
26	50	Targeted	70	71.26 [70.81, 71.70]	14.54 [14.03, 15.05]

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 , coverage L and outcomes are percentages. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. Targeted selection prioritizes accounts by degree. The response interval is $\tau_L = 5$ steps and TPR = TNR = 95%; 70% is the benchmark coverage. All networks are BA with $N = 1000$. Repeated condition IDs denote the same runs. Unrounded results, condition IDs and sample sizes are in `tableS3_P5_legal.csv`.

Table S5 Platform outcomes under random and degree-targeted selection

ID	p_0 (%)	Selection	n_p (%)	$\bar{E}$	$O(50)$
115	10	Random	5	25.48 [25.29, 25.68]	10.47 [10.05, 10.89]
112	10	Targeted	5	28.92 [28.35, 29.49]	4.70 [4.61, 4.80]
116	10	Random	10	37.41 [37.12, 37.71]	26.52 [26.02, 27.02]
113	10	Targeted	10	40.94 [40.21, 41.66]	10.49 [10.31, 10.66]
117	10	Random	20	58.26 [57.95, 58.57]	60.97 [60.43, 61.51]
114	10	Targeted	20	62.48 [61.66, 63.30]	21.14 [20.72, 21.57]
59	10	Random	75	99.63 [99.59, 99.67]	99.94 [99.91, 99.97]
35	10	Targeted	75	99.73 [99.71, 99.75]	99.74 [99.67, 99.82]
127	30	Random	5	20.22 [20.05, 20.38]	7.85 [7.58, 8.12]
124	30	Targeted	5	16.52 [16.26, 16.77]	4.28 [4.17, 4.38]
128	30	Random	10	32.92 [32.69, 33.15]	22.34 [21.90, 22.78]
125	30	Targeted	10	25.81 [25.47, 26.16]	10.01 [9.88, 10.15]
129	30	Random	20	54.83 [54.57, 55.10]	57.35 [56.79, 57.91]
126	30	Targeted	20	42.75 [42.12, 43.38]	20.37 [20.08, 20.66]
61	30	Random	75	99.15 [99.08, 99.22]	99.96 [99.93, 99.98]
44	30	Targeted	75	99.26 [99.23, 99.30]	99.65 [99.59, 99.72]
139	50	Random	5	17.79 [17.62, 17.96]	6.70 [6.44, 6.96]
136	50	Targeted	5	10.96 [10.78, 11.13]	4.01 [3.88, 4.13]
140	50	Random	10	30.46 [30.31, 30.60]	20.94 [20.47, 21.41]
137	50	Targeted	10	18.25 [18.03, 18.47]	9.67 [9.53, 9.81]
141	50	Random	20	52.71 [52.51, 52.92]	54.81 [54.41, 55.22]
138	50	Targeted	20	31.88 [31.52, 32.23]	20.07 [19.83, 20.31]
63	50	Random	75	98.80 [98.74, 98.85]	99.95 [99.92, 99.97]
53	50	Targeted	75	98.84 [98.81, 98.88]	99.68 [99.63, 99.74]

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 , per-step coverage n_p and outcomes are percentages. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. Targeted selection prioritizes accounts by degree. Workload is fixed and TPR = TNR = 80%; 75% is the benchmark coverage. All networks are BA with $N = 1000$. Repeated condition IDs denote the same runs. Unrounded results, condition IDs and sample sizes are in `tableS3_P5_platform.csv`.

Table S6 Legal outcomes across true-positive and true-negative rates

ID	p_0 (%)	TPR (%)	TNR (%)	$\bar{E}$	$O(50)$	$P_R(50)$	n_{PR}
64	10	90	90	72.55 [72.03, 73.06]	30.04 [29.43, 30.65]	88.84 [88.52, 89.17]	30
142	10	90	95	72.55 [72.05, 73.05]	16.73 [16.37, 17.08]	94.00 [93.80, 94.20]	30
143	10	90	100	72.60 [72.00, 73.19]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
144	10	95	90	77.90 [77.25, 78.55]	33.41 [32.81, 34.02]	86.50 [85.99, 87.01]	30
8	10	95	95	77.86 [77.39, 78.34]	18.39 [17.98, 18.81]	92.94 [92.72, 93.15]	30
145	10	95	100	78.07 [77.50, 78.64]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
146	10	100	90	83.25 [82.69, 83.81]	35.76 [35.09, 36.43]	83.78 [83.24, 84.33]	30
147	10	100	95	83.36 [82.82, 83.90]	20.49 [20.00, 20.98]	90.99 [90.60, 91.38]	30
65	10	100	100	83.52 [82.72, 84.33]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
66	30	90	90	68.15 [67.72, 68.57]	26.44 [25.88, 27.00]	91.42 [91.16, 91.68]	30
148	30	90	95	68.59 [68.09, 69.10]	14.53 [14.19, 14.86]	95.51 [95.38, 95.63]	30
149	30	90	100	68.47 [68.00, 68.94]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
150	30	95	90	73.75 [73.16, 74.35]	29.42 [28.71, 30.12]	89.89 [89.53, 90.25]	30
17	30	95	95	73.59 [73.19, 74.00]	16.18 [15.83, 16.53]	94.74 [94.61, 94.87]	30
151	30	95	100	73.81 [73.33, 74.30]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
152	30	100	90	79.53 [78.98, 80.08]	32.48 [31.80, 33.16]	87.78 [87.39, 88.17]	30
153	30	100	95	78.99 [78.41, 79.57]	18.02 [17.53, 18.51]	93.60 [93.32, 93.87]	30
67	30	100	100	79.30 [78.81, 79.79]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
68	50	90	90	66.30 [65.86, 66.74]	24.30 [23.65, 24.96]	92.66 [92.43, 92.90]	30
154	50	90	95	65.87 [65.41, 66.33]	13.07 [12.74, 13.41]	96.27 [96.15, 96.38]	30
155	50	90	100	66.09 [65.67, 66.51]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
156	50	95	90	71.79 [71.22, 72.36]	27.49 [26.89, 28.10]	91.21 [90.89, 91.52]	30
26	50	95	95	71.26 [70.81, 71.70]	14.54 [14.03, 15.05]	95.67 [95.49, 95.86]	30
157	50	95	100	71.12 [70.62, 71.62]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30
158	50	100	90	76.43 [75.92, 76.93]	29.36 [28.88, 29.84]	90.25 [89.98, 90.53]	30
159	50	100	95	76.58 [76.14, 77.02]	16.29 [15.89, 16.69]	94.86 [94.71, 95.01]	30
69	50	100	100	76.83 [76.40, 77.26]	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	30

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 , TPR, TNR and outcomes are percentages. TPR and TNR are the true-positive and true-negative rates. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios. n_{PR} is the number of runs with defined precision (30 in every condition). Legal coverage is $L = 70\%$, the response interval is $\tau_L = 5$ steps, and selection is degree-targeted. All networks are BA with $N = 1000$. Unrounded results, condition IDs and sample sizes are in `tableS4_P6_legal.csv`.

Table S7 Platform outcomes across true-positive and true-negative rates at 50% coverage

ID	p_0 (%)	TPR (%)	TNR (%)	$\bar{E}$	$O(50)$	$P_R(50)$	n_{PR}
31	10	70	70	84.56 [83.82, 85.30]	53.99 [52.48, 55.49]	20.12 [19.55, 20.68]	30
160	10	70	80	85.06 [84.11, 86.01]	53.76 [52.16, 55.36]	26.99 [26.04, 27.93]	30
161	10	70	90	84.47 [83.74, 85.19]	50.54 [50.03, 51.04]	42.62 [41.73, 43.51]	30
162	10	80	70	90.15 [89.46, 90.84]	55.97 [53.60, 58.34]	13.76 [13.07, 14.46]	30
32	10	80	80	89.59 [88.69, 90.49]	53.00 [51.64, 54.37]	19.65 [18.44, 20.86]	30
163	10	80	90	89.40 [88.57, 90.23]	51.48 [50.79, 52.16]	33.70 [32.46, 34.95]	30
164	10	90	70	92.75 [92.12, 93.38]	54.23 [51.92, 56.53]	10.11 [9.46, 10.76]	30
165	10	90	80	92.52 [92.01, 93.04]	51.91 [51.12, 52.70]	14.55 [13.81, 15.28]	30
33	10	90	90	93.32 [92.65, 93.99]	52.11 [51.44, 52.77]	24.15 [22.83, 25.47]	30
40	30	70	70	72.12 [70.96, 73.29]	52.56 [51.25, 53.86]	31.35 [30.75, 31.96]	30
172	30	70	80	72.31 [71.14, 73.48]	51.79 [50.93, 52.65]	40.42 [39.80, 41.04]	30
173	30	70	90	71.77 [70.83, 72.71]	49.76 [49.42, 50.10]	57.91 [57.29, 58.52]	30
174	30	80	70	76.91 [75.71, 78.11]	52.72 [51.08, 54.35]	26.66 [25.93, 27.40]	30
41	30	80	80	77.76 [76.42, 79.10]	52.50 [51.47, 53.53]	35.04 [34.05, 36.03]	30
175	30	80	90	77.14 [75.79, 78.48]	50.45 [49.92, 50.97]	52.07 [50.97, 53.17]	30
176	30	90	70	82.33 [81.17, 83.49]	53.85 [52.46, 55.23]	21.86 [21.07, 22.65]	30
177	30	90	80	80.42 [79.66, 81.18]	51.37 [50.81, 51.93]	31.10 [30.47, 31.73]	30
42	30	90	90	81.24 [80.18, 82.29]	50.64 [50.05, 51.23]	46.58 [45.41, 47.75]	30
49	50	70	70	62.93 [62.06, 63.80]	51.31 [50.79, 51.83]	37.84 [37.47, 38.20]	30
184	50	70	80	64.56 [63.05, 66.07]	51.57 [50.79, 52.35]	47.31 [46.63, 47.98]	30
185	50	70	90	62.46 [61.42, 63.50]	49.37 [49.10, 49.65]	64.94 [64.46, 65.42]	30
186	50	80	70	68.18 [66.48, 69.88]	52.58 [51.28, 53.89]	33.85 [32.93, 34.78]	30
50	50	80	80	68.42 [66.92, 69.92]	51.44 [50.71, 52.16]	43.29 [42.40, 44.18]	30
187	50	80	90	69.72 [67.86, 71.58]	50.19 [49.67, 50.71]	59.96 [58.81, 61.11]	30
188	50	90	70	72.87 [71.06, 74.68]	53.22 [51.82, 54.62]	30.19 [29.22, 31.16]	30
189	50	90	80	73.35 [71.62, 75.07]	52.37 [51.46, 53.29]	38.83 [37.79, 39.88]	30
51	50	90	90	72.03 [70.39, 73.68]	50.03 [49.61, 50.44]	56.68 [55.58, 57.79]	30

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 , TPR, TNR and outcomes are percentages. TPR and TNR are the true-positive and true-negative rates. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios. n_{PR} is the number of runs with defined precision (30 in every condition). Per-step audit coverage is fixed at $n_p = 50\%$ and selection is degree-targeted. All networks are BA with $N = 1000$. Unrounded results, condition IDs and sample sizes are in `tableS4_P6_platform50.csv`.

Table S8 Platform outcomes across true-positive and true-negative rates at 75% coverage

ID	p_0 (%)	TPR (%)	TNR (%)	$\bar{E}$	$O(50)$	$P_R(50)$	n_{PR}
34	10	70	70	99.52 [99.46, 99.59]	99.97 [99.95, 100.00]	2.18 [1.91, 2.45]	30
166	10	70	80	99.52 [99.48, 99.56]	99.72 [99.65, 99.79]	3.26 [3.01, 3.50]	30
167	10	70	90	99.50 [99.45, 99.55]	95.79 [95.56, 96.02]	6.50 [5.94, 7.06]	30
168	10	80	70	99.74 [99.72, 99.75]	99.99 [99.97, 100.00]	1.68 [1.59, 1.76]	30
35	10	80	80	99.73 [99.71, 99.75]	99.74 [99.67, 99.82]	2.44 [2.32, 2.56]	30
169	10	80	90	99.73 [99.72, 99.75]	95.77 [95.54, 96.00]	4.79 [4.62, 4.96]	30
170	10	90	70	99.82 [99.80, 99.83]	99.98 [99.96, 100.00]	1.46 [1.41, 1.52]	30
171	10	90	80	99.82 [99.81, 99.84]	99.74 [99.67, 99.80]	2.11 [2.04, 2.18]	30
36	10	90	90	99.83 [99.82, 99.84]	95.77 [95.54, 95.99]	4.19 [4.05, 4.34]	30
43	30	70	70	98.82 [98.76, 98.87]	99.97 [99.95, 99.99]	5.22 [5.06, 5.38]	30
178	30	70	80	98.85 [98.79, 98.92]	99.66 [99.58, 99.73]	7.48 [7.16, 7.80]	30
179	30	70	90	98.76 [98.70, 98.83]	95.86 [95.61, 96.12]	14.65 [14.10, 15.21]	30
180	30	80	70	99.27 [99.24, 99.30]	99.98 [99.96, 100.00]	4.30 [4.19, 4.41]	30
44	30	80	80	99.26 [99.23, 99.30]	99.65 [99.59, 99.72]	6.39 [6.21, 6.57]	30
181	30	80	90	99.26 [99.24, 99.29]	95.59 [95.37, 95.80]	12.18 [11.93, 12.43]	30
182	30	90	70	99.49 [99.47, 99.51]	99.99 [99.98, 100.00]	3.84 [3.78, 3.90]	30
183	30	90	80	99.49 [99.47, 99.51]	99.71 [99.66, 99.77]	5.73 [5.63, 5.83]	30
45	30	90	90	99.50 [99.48, 99.52]	95.42 [95.16, 95.68]	10.76 [10.58, 10.94]	30
52	50	70	70	98.23 [98.15, 98.30]	99.99 [99.97, 100.00]	7.68 [7.41, 7.95]	30
190	50	70	80	98.27 [98.21, 98.33]	99.64 [99.56, 99.72]	10.74 [10.50, 10.97]	30
191	50	70	90	98.29 [98.23, 98.34]	95.24 [94.99, 95.50]	19.34 [18.88, 19.79]	30
192	50	80	70	98.84 [98.80, 98.89]	99.98 [99.96, 100.00]	6.51 [6.41, 6.62]	30
53	50	80	80	98.84 [98.81, 98.88]	99.68 [99.63, 99.74]	9.52 [9.37, 9.68]	30
193	50	80	90	98.84 [98.81, 98.88]	95.56 [95.29, 95.83]	17.27 [17.02, 17.52]	30
194	50	90	70	99.18 [99.15, 99.21]	99.98 [99.96, 100.00]	5.90 [5.84, 5.97]	30
195	50	90	80	99.20 [99.18, 99.23]	99.65 [99.58, 99.73]	8.64 [8.53, 8.74]	30
54	50	90	90	99.22 [99.19, 99.24]	95.49 [95.19, 95.79]	15.73 [15.59, 15.87]	30

Note. Means with pointwise 95% Student- t confidence intervals in brackets; 30 independent runs per condition. Intervals are not clipped. p_0 , TPR, TNR and outcomes are percentages. TPR and TNR are the true-positive and true-negative rates. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios. n_{PR} is the number of runs with defined precision (30 in every condition). Per-step audit coverage is fixed at $n_p = 75\%$ and selection is degree-targeted. All networks are BA with $N = 1000$. Unrounded results, condition IDs and sample sizes are in `tableS4_P6_platform75.csv`.

Table S9 H1: differences in mean-effectiveness gains

Regime	p_0 (%)	Difference in gains	$\Delta\Delta\bar{E}$ (pp)
Legal	10	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 5$	-34.01 [-35.09, -32.92]
Legal	10	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 10$	-26.48 [-27.42, -25.54]
Platform	10	Accuracy gain at $n_p = 100$ minus gain at $n_p = 50$	-8.62 [-9.60, -7.64]
Platform	10	Accuracy gain at $n_p = 100$ minus gain at $n_p = 75$	-0.17 [-0.24, -0.10]
Legal	30	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 5$	-29.58 [-30.83, -28.33]
Legal	30	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 10$	-18.95 [-20.18, -17.72]
Platform	30	Accuracy gain at $n_p = 100$ minus gain at $n_p = 50$	-8.76 [-10.30, -7.22]
Platform	30	Accuracy gain at $n_p = 100$ minus gain at $n_p = 75$	-0.33 [-0.39, -0.27]
Legal	50	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 5$	-25.71 [-27.75, -23.68]
Legal	50	Coverage gain at $\tau_L = 1$ minus gain at $\tau_L = 10$	-11.73 [-13.74, -9.71]
Platform	50	Accuracy gain at $n_p = 100$ minus gain at $n_p = 50$	-8.59 [-10.42, -6.76]
Platform	50	Accuracy gain at $n_p = 100$ minus gain at $n_p = 75$	-0.48 [-0.56, -0.40]

Note. Entries are differences in gains formed from four independent condition means, in percentage points, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. Legal coverage gain is $\bar{E}(L = 90\%) - \bar{E}(L = 50\%)$ at the stated response interval, with TPR = TNR = 95%. Platform accuracy gain is $\bar{E}(\text{TPR} = \text{TNR} = 90\%) - \bar{E}(\text{TPR} = \text{TNR} = 70\%)$ at the stated coverage. $\bar{E}$ averages steps 1–50. Coverage values in the row labels are percentages; response intervals are steps. The 12 contrasts concern the specified endpoints; contrasts sharing conditions reuse the same runs. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS9_H1_source.csv`.

Table S10 H2: PAWN index differences and wrongful-exposure comparisons

Regime	p_0 (%)	Index difference	Estimate	Row-bootstrap 95% interval	Within-point MC 95% interval
A. Mean effectiveness $\bar{E}$: primary H2 comparisons					
Legal	10	L minus α_L	0.318	[0.237, 0.351]	[0.315, 0.325]
Legal	10	τ_L minus α_L	0.123	[0.050, 0.184]	[0.118, 0.132]
Platform	10	n_p minus α_P	0.442	[0.373, 0.461]	[0.439, 0.454]
Legal	30	L minus α_L	0.339	[0.259, 0.369]	[0.336, 0.344]
Legal	30	τ_L minus α_L	0.114	[0.040, 0.168]	[0.110, 0.120]
Platform	30	n_p minus α_P	0.490	[0.412, 0.504]	[0.486, 0.495]
Legal	50	L minus α_L	0.360	[0.276, 0.379]	[0.355, 0.362]
Legal	50	τ_L minus α_L	0.116	[0.037, 0.161]	[0.111, 0.120]
Platform	50	n_p minus α_P	0.503	[0.429, 0.519]	[0.498, 0.508]
B. Wrongful exposure $O(50)$: companion comparisons					
Legal	10	L minus α_L	0.077	[0.007, 0.154]	[0.066, 0.088]
Legal	10	τ_L minus α_L	0.012	[−0.050, 0.106]	[0.001, 0.025]
Platform	10	n_p minus α_P	0.344	[0.295, 0.413]	[0.341, 0.353]
Legal	30	L minus α_L	0.108	[0.017, 0.172]	[0.098, 0.117]
Legal	30	τ_L minus α_L	0.051	[−0.021, 0.117]	[0.042, 0.059]
Platform	30	n_p minus α_P	0.359	[0.303, 0.418]	[0.350, 0.370]
Legal	50	L minus α_L	0.121	[0.042, 0.185]	[0.110, 0.133]
Legal	50	τ_L minus α_L	0.069	[−0.004, 0.143]	[0.057, 0.081]
Platform	50	n_p minus α_P	0.380	[0.318, 0.419]	[0.361, 0.380]

Note. Entries are dimensionless differences in median PAWN indices across 10 fixed input bins, not percentage points or variance shares. The nine $\bar{E}$ comparisons test H2; the nine $O(50)$ comparisons are companion analyses. Each regime– p_0 stratum uses 500 Latin hypercube points and 30 independent runs per point, with symmetric accuracy (TPR = TNR). Both interval types are marginal 95% intervals based on 1,000 resamples each. Row bootstrap measures approximate stability under design-point resampling, which does not preserve Latin hypercube stratification; within-point bootstrap measures Monte Carlo uncertainty at fixed parameter points. These separate intervals are not combined. Unrounded estimates, interval endpoints, input names, comparison IDs and valid resample counts are in `tableS10_H2_source.csv`.

Table S11 H3: effectiveness and wrongful-exposure changes

p_0 (%)	Path and subtraction direction	$\Delta \bar{E}$ (pp)	$\Delta O(50)$ (pp)
10	Legal coverage: $L = 90$ minus 50 ($\tau_L = 5$)	39.92 [38.95, 40.89]	21.03 [20.25, 21.81]
10	Legal response interval: $\tau_L = 1$ minus 10 ($L = 70$)	49.76 [49.13, 50.39]	72.36 [71.76, 72.96]
10	Platform audit coverage: $n_p = 100$ minus 50 ($\alpha_P = 80$)	10.33 [9.43, 11.23]	47.00 [45.63, 48.36]
10	Legal accuracy: $\alpha_L = 100$ minus 90 ($L = 70, \tau_L = 5$)	10.98 [10.04, 11.91]	-30.04 [-30.65, -29.43]
10	Platform accuracy: $\alpha_P = 90$ minus 70 ($n_p = 75$)	0.30 [0.24, 0.37]	-4.21 [-4.43, -3.98]
30	Legal coverage: $L = 90$ minus 50 ($\tau_L = 5$)	46.19 [45.50, 46.88]	20.52 [19.84, 21.20]
30	Legal response interval: $\tau_L = 1$ minus 10 ($L = 70$)	54.57 [54.23, 54.91]	73.25 [72.71, 73.79]
30	Platform audit coverage: $n_p = 100$ minus 50 ($\alpha_P = 80$)	22.03 [20.69, 23.37]	47.50 [46.47, 48.53]
30	Legal accuracy: $\alpha_L = 100$ minus 90 ($L = 70, \tau_L = 5$)	11.15 [10.52, 11.78]	-26.44 [-27.00, -25.88]
30	Platform accuracy: $\alpha_P = 90$ minus 70 ($n_p = 75$)	0.68 [0.63, 0.74]	-4.55 [-4.81, -4.29]
50	Legal coverage: $L = 90$ minus 50 ($\tau_L = 5$)	50.99 [50.42, 51.56]	20.31 [19.54, 21.08]
50	Legal response interval: $\tau_L = 1$ minus 10 ($L = 70$)	56.62 [56.31, 56.94]	74.17 [73.69, 74.64]
50	Platform audit coverage: $n_p = 100$ minus 50 ($\alpha_P = 80$)	31.24 [29.74, 32.75]	48.56 [47.84, 49.29]
50	Legal accuracy: $\alpha_L = 100$ minus 90 ($L = 70, \tau_L = 5$)	10.53 [9.93, 11.14]	-24.30 [-24.96, -23.65]
50	Platform accuracy: $\alpha_P = 90$ minus 70 ($n_p = 75$)	0.99 [0.92, 1.07]	-4.50 [-4.80, -4.20]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. Five paths are evaluated at each p_0 (15 comparisons). Subtraction follows the stated direction. Coverage and accuracy levels are percentages; response intervals are steps. Accuracy paths change TPR and TNR together; legal capacity paths hold both at 95%. The separate outcome intervals do not form a joint 95% confidence region. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS11_H3_source.csv`.

Table S12 Outcome differences: state-responsive minus fixed moderation

p_0 (%)	κ	$\Delta \bar{E}$ (pp)	$\Delta O(50)$ (pp)	$\Delta P_R(50)$ (pp)
10	1	-24.73 [-24.98, -24.47]	-59.92 [-60.69, -59.14]	44.33 [44.05, 44.62]
10	2	-15.63 [-15.92, -15.35]	-49.82 [-50.22, -49.43]	29.77 [29.46, 30.07]
10	5	-7.15 [-7.26, -7.04]	-39.99 [-40.77, -39.21]	15.45 [15.18, 15.72]
30	1	-29.85 [-30.07, -29.63]	-51.87 [-52.25, -51.48]	42.84 [42.53, 43.15]
30	2	-17.67 [-17.88, -17.47]	-44.97 [-45.35, -44.59]	27.83 [27.53, 28.12]
30	5	-7.31 [-7.40, -7.23]	-34.02 [-34.82, -33.22]	15.20 [14.86, 15.53]
50	1	-32.32 [-32.58, -32.07]	-49.49 [-49.90, -49.08]	41.07 [40.78, 41.36]
50	2	-19.00 [-19.16, -18.84]	-43.89 [-44.38, -43.40]	26.97 [26.73, 27.21]
50	5	-7.13 [-7.21, -7.05]	-32.87 [-33.73, -32.02]	15.21 [14.91, 15.51]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios. Subtraction is responsive moderation at the stated κ minus fixed moderation. Fixed per-step coverage is 75%, also the responsive ceiling; both modes use degree targeting, TPR = TNR = 80%, and the whole network as the candidate pool. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS12_P3_differences_source.csv`.

Table S13 Outcome differences: degree-targeted minus random selection

Regime	p_0 (%)	Coverage (%)	Role	$\Delta \bar{E}$ (pp)	$\Delta O(50)$ (pp)
Legal	10	5	Low coverage	0.14 [−0.17, 0.46]	−0.23 [−0.32, −0.14]
Legal	10	10	Low coverage	−0.19 [−0.64, 0.25]	−0.42 [−0.57, −0.26]
Legal	10	20	Low coverage	0.36 [−0.10, 0.82]	−0.31 [−0.51, −0.10]
Legal	10	70	Benchmark	10.63 [10.00, 11.27]	2.95 [2.32, 3.59]
Platform	10	5	Low coverage	3.44 [2.84, 4.03]	−5.77 [−6.20, −5.33]
Platform	10	10	Low coverage	3.52 [2.75, 4.30]	−16.04 [−16.56, −15.51]
Platform	10	20	Low coverage	4.22 [3.36, 5.09]	−39.83 [−40.50, −39.15]
Platform	10	75	Benchmark	0.10 [0.06, 0.14]	−0.20 [−0.28, −0.12]
Legal	30	5	Low coverage	−0.93 [−1.10, −0.76]	−0.15 [−0.23, −0.07]
Legal	30	10	Low coverage	−1.93 [−2.14, −1.72]	−0.32 [−0.43, −0.21]
Legal	30	20	Low coverage	−2.81 [−3.07, −2.56]	−0.57 [−0.73, −0.41]
Legal	30	70	Benchmark	9.07 [8.59, 9.55]	2.55 [2.07, 3.03]
Platform	30	5	Low coverage	−3.70 [−4.00, −3.40]	−3.58 [−3.86, −3.29]
Platform	30	10	Low coverage	−7.10 [−7.51, −6.70]	−12.32 [−12.78, −11.87]
Platform	30	20	Low coverage	−12.08 [−12.76, −11.41]	−36.98 [−37.60, −36.36]
Platform	30	75	Benchmark	0.12 [0.04, 0.19]	−0.30 [−0.37, −0.23]
Legal	50	5	Low coverage	−1.44 [−1.60, −1.29]	−0.14 [−0.20, −0.08]
Legal	50	10	Low coverage	−3.02 [−3.15, −2.89]	−0.24 [−0.33, −0.14]
Legal	50	20	Low coverage	−4.48 [−4.68, −4.28]	−0.27 [−0.43, −0.11]
Legal	50	70	Benchmark	8.30 [7.80, 8.80]	2.41 [1.78, 3.04]
Platform	50	5	Low coverage	−6.83 [−7.07, −6.60]	−2.69 [−2.98, −2.40]
Platform	50	10	Low coverage	−12.20 [−12.47, −11.94]	−11.27 [−11.76, −10.78]
Platform	50	20	Low coverage	−20.84 [−21.25, −20.43]	−34.74 [−35.20, −34.28]
Platform	50	75	Benchmark	0.05 [−0.02, 0.11]	−0.26 [−0.32, −0.20]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. Coverage is L for legal governance and per-step n_p for platform governance. Legal settings are $\tau_L = 5$ and $\text{TPR} = \text{TNR} = 95\%$; platform workload is fixed with $\text{TPR} = \text{TNR} = 80\%$. The table retains 18 low-coverage comparisons and six benchmark references. Equal coverage percentages do not imply equal cumulative review budgets; intervals crossing zero do not establish equivalence. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS13_P5_differences_source.csv`.

Table S14 Outcome differences from separately increasing TPR and TNR

Regime	p_0 (%)	Coverage (%)	Varied rate (%)	Fixed rate (%)	$\Delta \bar{E}$ (pp)	$\Delta O(50)$ (pp)	$\Delta P_R(50)$ (pp)
Legal	10	70	TPR: 100 – 90	TNR = 90	10.71 [9.96, 11.45]	5.72 [4.83, 6.61]	–5.06 [–5.68, –4.44]
Legal	10	70	TPR: 100 – 90	TNR = 95	10.81 [10.09, 11.53]	3.77 [3.17, 4.36]	–3.01 [–3.44, –2.57]
Legal	10	70	TPR: 100 – 90	TNR = 100	10.93 [9.95, 11.90]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Legal	10	70	TNR: 100 – 90	TPR = 90	0.05 [–0.72, 0.82]	–30.04 [–30.65, –29.43]	11.16 [10.83, 11.48]
Legal	10	70	TNR: 100 – 90	TPR = 95	0.17 [–0.68, 1.02]	–33.41 [–34.02, –32.81]	13.50 [12.99, 14.01]
Legal	10	70	TNR: 100 – 90	TPR = 100	0.27 [–0.69, 1.23]	–35.76 [–36.43, –35.09]	16.22 [15.67, 16.76]
Platform	10	50	TPR: 90 – 70	TNR = 70	8.19 [7.24, 9.15]	0.24 [–2.46, 2.94]	–10.01 [–10.85, –9.17]
Platform	10	50	TPR: 90 – 70	TNR = 80	7.47 [6.40, 8.53]	–1.85 [–3.61, –0.09]	–12.44 [–13.61, –11.27]
Platform	10	50	TPR: 90 – 70	TNR = 90	8.85 [7.89, 9.82]	1.57 [0.75, 2.39]	–18.47 [–20.04, –16.91]
Platform	10	50	TNR: 90 – 70	TPR = 70	–0.09 [–1.11, 0.92]	–3.45 [–5.02, –1.88]	22.50 [21.46, 23.54]
Platform	10	50	TNR: 90 – 70	TPR = 80	–0.75 [–1.81, 0.31]	–4.50 [–6.95, –2.04]	19.94 [18.53, 21.34]
Platform	10	50	TNR: 90 – 70	TPR = 90	0.57 [–0.33, 1.47]	–2.12 [–4.50, 0.26]	14.04 [12.59, 15.49]
Platform	10	75	TPR: 90 – 70	TNR = 70	0.29 [0.23, 0.36]	0.01 [–0.02, 0.03]	–0.72 [–0.99, –0.44]
Platform	10	75	TPR: 90 – 70	TNR = 80	0.31 [0.27, 0.35]	0.02 [–0.07, 0.11]	–1.15 [–1.40, –0.89]
Platform	10	75	TPR: 90 – 70	TNR = 90	0.33 [0.27, 0.38]	–0.02 [–0.34, 0.29]	–2.31 [–2.88, –1.74]
Platform	10	75	TNR: 90 – 70	TPR = 70	–0.02 [–0.10, 0.06]	–4.18 [–4.41, –3.95]	4.32 [3.71, 4.93]
Platform	10	75	TNR: 90 – 70	TPR = 80	–0.01 [–0.03, 0.02]	–4.22 [–4.45, –3.99]	3.11 [2.93, 3.30]
Platform	10	75	TNR: 90 – 70	TPR = 90	0.01 [–0.01, 0.03]	–4.21 [–4.44, –3.99]	2.73 [2.58, 2.88]
Legal	30	70	TPR: 100 – 90	TNR = 90	11.38 [10.70, 12.07]	6.04 [5.18, 6.90]	–3.64 [–4.10, –3.18]
Legal	30	70	TPR: 100 – 90	TNR = 95	10.40 [9.65, 11.15]	3.49 [2.91, 4.07]	–1.91 [–2.21, –1.61]
Legal	30	70	TPR: 100 – 90	TNR = 100	10.83 [10.16, 11.50]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Legal	30	70	TNR: 100 – 90	TPR = 90	0.32 [–0.30, 0.94]	–26.44 [–27.00, –25.88]	8.58 [8.32, 8.84]
Legal	30	70	TNR: 100 – 90	TPR = 95	0.06 [–0.69, 0.81]	–29.42 [–30.12, –28.71]	10.11 [9.75, 10.47]
Legal	30	70	TNR: 100 – 90	TPR = 100	–0.23 [–0.96, 0.49]	–32.48 [–33.16, –31.80]	12.22 [11.83, 12.61]
Platform	30	50	TPR: 90 – 70	TNR = 70	10.21 [8.60, 11.81]	1.29 [–0.57, 3.15]	–9.49 [–10.47, –8.52]
Platform	30	50	TPR: 90 – 70	TNR = 80	8.12 [6.75, 9.49]	–0.42 [–1.43, 0.59]	–9.32 [–10.18, –8.45]
Platform	30	50	TPR: 90 – 70	TNR = 90	9.47 [8.08, 10.85]	0.88 [0.21, 1.55]	–11.32 [–12.63, –10.02]
Platform	30	50	TNR: 90 – 70	TPR = 70	–0.36 [–1.82, 1.11]	–2.80 [–4.14, –1.46]	26.55 [25.71, 27.40]
Platform	30	50	TNR: 90 – 70	TPR = 80	0.23 [–1.54, 2.00]	–2.27 [–3.98, –0.56]	25.41 [24.11, 26.70]
Platform	30	50	TNR: 90 – 70	TPR = 90	–1.09 [–2.62, 0.44]	–3.21 [–4.70, –1.72]	24.72 [23.33, 26.11]
Platform	30	75	TPR: 90 – 70	TNR = 70	0.68 [0.62, 0.73]	0.02 [0.00, 0.04]	–1.38 [–1.55, –1.21]
Platform	30	75	TPR: 90 – 70	TNR = 80	0.64 [0.57, 0.70]	0.06 [–0.04, 0.15]	–1.75 [–2.08, –1.42]
Platform	30	75	TPR: 90 – 70	TNR = 90	0.74 [0.67, 0.80]	–0.44 [–0.80, –0.09]	–3.90 [–4.48, –3.32]

Continued on next page

Table S14 Outcome differences from separately increasing TPR and TNR (continued)

Regime	p_0 (%)	Coverage (%)	Varied rate (%)	Fixed rate (%)	$\Delta \bar{E}$ (pp)	$\Delta O(50)$ (pp)	$\Delta P_R(50)$ (pp)
Platform	30	75	TNR: 90 – 70	TPR = 70	−0.05 [−0.14, 0.03]	−4.11 [−4.36, −3.85]	9.43 [8.86, 10.01]
Platform	30	75	TNR: 90 – 70	TPR = 80	−0.01 [−0.05, 0.03]	−4.39 [−4.61, −4.18]	7.88 [7.61, 8.15]
Platform	30	75	TNR: 90 – 70	TPR = 90	0.01 [−0.02, 0.03]	−4.57 [−4.83, −4.31]	6.92 [6.73, 7.10]
Legal	50	70	TPR: 100 – 90	TNR = 90	10.13 [9.47, 10.79]	5.06 [4.26, 5.85]	−2.41 [−2.77, −2.06]
Legal	50	70	TPR: 100 – 90	TNR = 95	10.71 [10.08, 11.33]	3.22 [2.71, 3.73]	−1.41 [−1.60, −1.22]
Legal	50	70	TPR: 100 – 90	TNR = 100	10.74 [10.15, 11.33]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Legal	50	70	TNR: 100 – 90	TPR = 90	−0.21 [−0.80, 0.39]	−24.30 [−24.96, −23.65]	7.34 [7.10, 7.57]
Legal	50	70	TNR: 100 – 90	TPR = 95	−0.67 [−1.41, 0.07]	−27.49 [−28.10, −26.89]	8.79 [8.48, 9.11]
Legal	50	70	TNR: 100 – 90	TPR = 100	0.40 [−0.25, 1.05]	−29.36 [−29.84, −28.88]	9.75 [9.47, 10.02]
Platform	50	50	TPR: 90 – 70	TNR = 70	9.94 [7.96, 11.92]	1.92 [0.44, 3.40]	−7.65 [−8.67, −6.62]
Platform	50	50	TPR: 90 – 70	TNR = 80	8.79 [6.54, 11.03]	0.80 [−0.38, 1.98]	−8.47 [−9.70, −7.25]
Platform	50	50	TPR: 90 – 70	TNR = 90	9.57 [7.66, 11.48]	0.65 [0.17, 1.14]	−8.26 [−9.45, −7.07]
Platform	50	50	TNR: 90 – 70	TPR = 70	−0.47 [−1.80, 0.86]	−1.93 [−2.51, −1.35]	27.10 [26.51, 27.70]
Platform	50	50	TNR: 90 – 70	TPR = 80	1.54 [−0.93, 4.00]	−2.40 [−3.79, −1.01]	26.11 [24.67, 27.56]
Platform	50	50	TNR: 90 – 70	TPR = 90	−0.83 [−3.22, 1.56]	−3.20 [−4.65, −1.75]	26.49 [25.06, 27.93]
Platform	50	75	TPR: 90 – 70	TNR = 70	0.95 [0.87, 1.03]	−0.01 [−0.03, 0.02]	−1.78 [−2.06, −1.50]
Platform	50	75	TPR: 90 – 70	TNR = 80	0.93 [0.87, 1.00]	0.01 [−0.09, 0.12]	−2.10 [−2.36, −1.84]
Platform	50	75	TPR: 90 – 70	TNR = 90	0.93 [0.87, 0.99]	0.25 [−0.14, 0.63]	−3.61 [−4.08, −3.14]
Platform	50	75	TNR: 90 – 70	TPR = 70	0.06 [−0.03, 0.15]	−4.74 [−5.00, −4.49]	11.66 [11.13, 12.18]
Platform	50	75	TNR: 90 – 70	TPR = 80	−0.00 [−0.06, 0.05]	−4.42 [−4.69, −4.15]	10.76 [10.49, 11.03]
Platform	50	75	TNR: 90 – 70	TPR = 90	0.04 [0.00, 0.08]	−4.49 [−4.79, −4.19]	9.82 [9.67, 9.98]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. TPR and TNR are the true-positive and true-negative rates. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $P_R(50)$ averages defined run-level precision ratios. Each contrast raises one rate while holding the other at the displayed value: legal 100% minus 90% at $L = 70\%$ and $\tau_L = 5$; platform 90% minus 70% at $n_p = 50\%$ or 75% with fixed workload. Selection is degree-targeted. The table contains 18 legal and 36 platform contrasts. Intervals crossing zero do not establish equivalence; TNR effects on pollution states are constrained by the model structure. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS14_P6_differences_source.csv`.

Table S15 Network-topology differences relative to BA

Regime	p_0 (%)	Difference	$\Delta \bar{E}$	$\Delta E(50)$	$\Delta O(50)$	$\Delta P_R(50)$
No governance	10	ER minus BA	4.44 [4.09, 4.80]	1.81 [1.65, 1.97]	0.00 [0.00, 0.00]	NA
No governance	10	WS minus BA	6.86 [6.48, 7.24]	0.02 [-0.00, 0.04]	0.00 [0.00, 0.00]	NA
Legal	10	ER minus BA	13.82 [13.17, 14.48]	25.98 [24.48, 27.49]	7.34 [6.64, 8.03]	-21.19 [-22.47, -19.91]
Legal	10	WS minus BA	10.50 [9.73, 11.28]	20.16 [18.53, 21.80]	7.03 [6.37, 7.69]	-15.74 [-16.92, -14.57]
Platform	10	ER minus BA	-2.76 [-3.00, -2.52]	-3.02 [-3.32, -2.72]	-21.59 [-22.51, -20.68]	1.78 [1.52, 2.05]
Platform	10	WS minus BA	-2.97 [-3.20, -2.73]	-3.48 [-3.79, -3.17]	-15.41 [-15.78, -15.03]	3.89 [3.61, 4.17]
No governance	30	ER minus BA	1.79 [1.52, 2.07]	1.42 [1.25, 1.59]	0.00 [0.00, 0.00]	NA
No governance	30	WS minus BA	0.53 [0.31, 0.74]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	NA
Legal	30	ER minus BA	6.12 [5.46, 6.77]	10.63 [9.20, 12.06]	5.24 [4.53, 5.94]	-6.46 [-7.04, -5.88]
Legal	30	WS minus BA	2.59 [1.81, 3.37]	5.43 [3.82, 7.05]	4.50 [3.86, 5.14]	-4.73 [-5.20, -4.25]
Platform	30	ER minus BA	-7.86 [-8.36, -7.35]	-8.39 [-9.11, -7.66]	-22.05 [-22.88, -21.22]	4.81 [4.29, 5.32]
Platform	30	WS minus BA	-7.74 [-8.10, -7.38]	-8.81 [-9.17, -8.45]	-15.44 [-15.90, -14.98]	9.12 [8.64, 9.60]
No governance	50	ER minus BA	0.82 [0.63, 1.01]	0.97 [0.86, 1.08]	0.00 [0.00, 0.00]	NA
No governance	50	WS minus BA	-0.58 [-0.68, -0.48]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	NA
Legal	50	ER minus BA	1.26 [0.69, 1.82]	3.09 [1.89, 4.30]	4.50 [3.84, 5.16]	-3.80 [-4.10, -3.50]
Legal	50	WS minus BA	-1.33 [-2.01, -0.66]	0.64 [-0.79, 2.06]	3.20 [2.54, 3.87]	-2.40 [-2.69, -2.11]
Platform	50	ER minus BA	-11.63 [-12.55, -10.72]	-12.21 [-13.55, -10.87]	-21.93 [-22.80, -21.06]	6.65 [5.94, 7.36]
Platform	50	WS minus BA	-10.95 [-11.38, -10.52]	-12.23 [-12.70, -11.75]	-15.37 [-15.72, -15.02]	12.29 [11.77, 12.81]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $E(50)$ is effectiveness at step 50; $P_R(50)$ averages defined run-level precision ratios. Precision is undefined without governance, so those differences are NA, not zero. Contrasts are ER minus BA and WS minus BA at $N = 1000$, with approximately matched mean degree near four. Networks retain isolates and disconnected realizations; other structural differences remain. Governance uses benchmark parameters and degree targeting. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS15_topology_differences_source.csv`.

Table S16 Network-size differences: 5,000 minus 1,000 nodes

Regime	p_0 (%)	Difference	$\Delta \bar{E}$	$\Delta E(50)$	$\Delta O(50)$	$\Delta P_R(50)$
No governance	10	5000 – 1000	−0.12 [−0.34, 0.11]	−0.00 [−0.01, 0.01]	0.00 [0.00, 0.00]	NA
Legal	10	5000 – 1000	−1.10 [−1.64, −0.56]	−1.64 [−2.83, −0.45]	−0.45 [−0.95, 0.05]	0.42 [0.16, 0.68]
Platform	10	5000 – 1000	−0.01 [−0.03, 0.01]	0.00 [0.00, 0.00]	−0.03 [−0.11, 0.04]	0.12 [−0.02, 0.27]
No governance	30	5000 – 1000	−0.07 [−0.21, 0.07]	0.00 [0.00, 0.01]	0.00 [0.00, 0.00]	NA
Legal	30	5000 – 1000	−0.48 [−0.92, −0.03]	−0.44 [−1.49, 0.61]	−0.34 [−0.73, 0.05]	0.19 [0.04, 0.34]
Platform	30	5000 – 1000	−0.01 [−0.05, 0.02]	0.00 [0.00, 0.00]	0.04 [−0.03, 0.11]	0.09 [−0.11, 0.29]
No governance	50	5000 – 1000	−0.06 [−0.14, 0.01]	0.00 [−0.00, 0.00]	0.00 [0.00, 0.00]	NA
Legal	50	5000 – 1000	−0.37 [−0.84, 0.10]	−1.22 [−2.32, −0.13]	0.06 [−0.48, 0.60]	0.01 [−0.18, 0.20]
Platform	50	5000 – 1000	−0.02 [−0.05, 0.02]	0.00 [0.00, 0.00]	0.00 [−0.06, 0.07]	0.05 [−0.13, 0.23]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. All differences are in percentage points. $\bar{E}$ averages effectiveness over steps 1–50; $O(50)$ is the percentage of distinct accounts ever wronged by step 50. $E(50)$ is effectiveness at step 50; $P_R(50)$ averages defined run-level precision ratios. Precision is undefined without governance, so those differences are NA, not zero. Each contrast is $N = 5000$ minus $N = 1000$ on BA networks with attachment parameter $m = 2$. Benchmark governance parameters, coverage percentages and response intervals remain fixed; selection is degree-targeted. Under governance, absolute review volumes therefore increase with N . Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS16_scale_differences_source.csv`.

Table S17 Review-burden differences: state-responsive minus fixed moderation

p_0 (%)	κ	Δ Reviews	Δ Clean reviews
10	1	-22893.27 [-23051.67, -22734.86]	-25300.17 [-25410.96, -25189.37]
10	2	-18350.40 [-18701.30, -17999.50]	-20140.27 [-20445.70, -19834.84]
10	5	-14110.17 [-14456.17, -13764.17]	-15085.13 [-15407.18, -14763.08]
30	1	-19382.97 [-19514.87, -19251.06]	-22289.30 [-22392.31, -22186.29]
30	2	-14938.40 [-15188.78, -14688.02]	-16912.73 [-17132.48, -16692.98]
30	5	-12228.67 [-12445.28, -12012.05]	-13236.13 [-13443.50, -13028.77]
50	1	-17504.47 [-17655.66, -17353.27]	-20613.87 [-20736.97, -20490.77]
50	2	-12822.80 [-13022.94, -12622.66]	-14948.17 [-15118.29, -14778.04]
50	5	-11969.83 [-12195.81, -11743.85]	-12937.70 [-13142.15, -12733.25]

Note. Differences between independent condition means, with marginal 95% Welch confidence intervals in brackets; 30 independent runs per condition and no multiplicity adjustment. Both differences are cumulative review-event counts over steps 1–50, including repeated reviews of an account. The Clean reviews column counts events involving accounts that were clean when reviewed (TN+FP). Neither outcome is a percentage or a count of distinct accounts. Subtraction is responsive moderation at the stated κ minus fixed moderation. Fixed per-step coverage is 75%, also the responsive ceiling; both modes use degree targeting, TPR = TNR = 80%, and the whole network as the candidate pool. Unrounded results, comparison IDs, condition weights and sample sizes are in `tableS17_P3_workload_differences_source.csv`.

2 Supplementary figures

Reading the figures

Mean effectiveness $\bar{E}$ averages steps 1–50 within each run. Wrongful exposure $O(50)$ is the percentage of distinct accounts ever wronged by step 50; removal precision $P_R(50)$ averages defined run-level ratios. Review totals count events, including repeated reviews of the same account. Figure S1 expresses review totals in thousands.

Figures S1–S4 summarise 30 independent runs per condition. Figures S5–S11 use 500 Latin hypercube sampling (LHS) points per governance regime and initial pollution level, with 30 runs per point. The same responses are reused across diagnostics and input–response panels.

Displayed 95% intervals are marginal, unadjusted and not clipped. Row-bootstrap intervals describe stability under design-point resampling; within-point Monte Carlo intervals describe simulation uncertainty at fixed parameter points. These separate bootstrap intervals are not combined and need not contain the point estimate.

Input–response lines connect descriptive bin means. Continuous inputs use fixed empirical-quantile bins from the full LHS design; legal response intervals use the pairs 1–2, 3–4, $\dots$, 19–20 steps. Full values and identifiers remain in the CSV files.

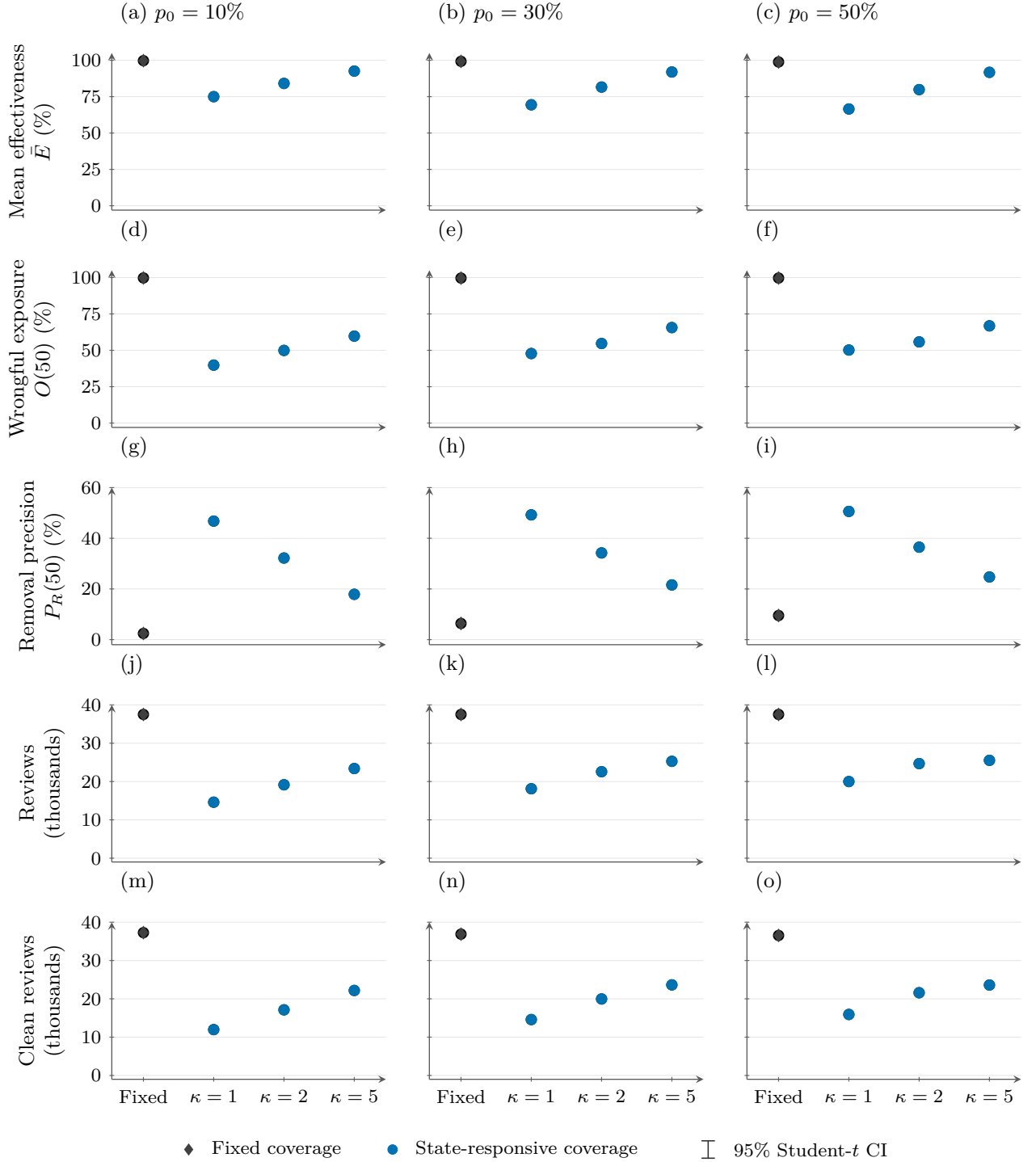


Figure S1: Platform workload specifications (P3). Fixed auditing reviews 75% of accounts per step; responsive auditing uses $\min\{N, \text{round}(0.75N), \text{round}(\kappa I_t)\}$, where I_t is the number of polluted accounts before review and $\kappa = 1, 2, 5$. Both modes use TPR=TNR= 80%. Points and whiskers show means and 95% Student- t CIs. Precision is defined in all 30 runs per condition; review counts are in thousands.

$$p_0 = 10\%$$

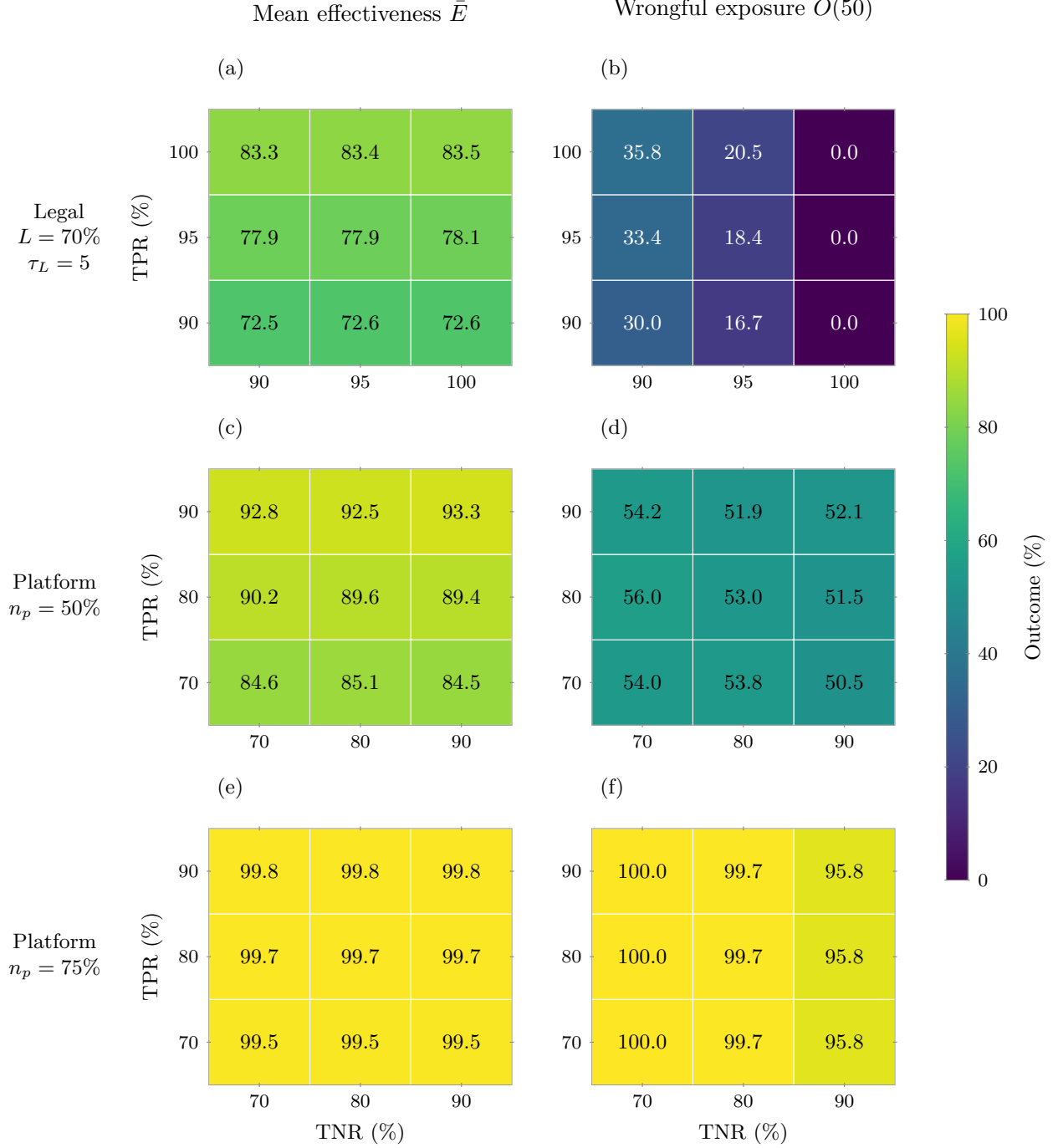


Figure S2: Independent TPR and TNR variation (P6), $p_0 = 10\%$. Rows identify governance and coverage settings; platform auditing is fixed. Cells show 30-run means, rounded to one decimal, on a common 0–100% scale. Full means and 95% Student- t CIs are retained in the plot CSVs. False-positive flags accumulate wrongful exposure without deleting accounts or links or feeding back into diffusion.

$$p_0 = 30\%$$

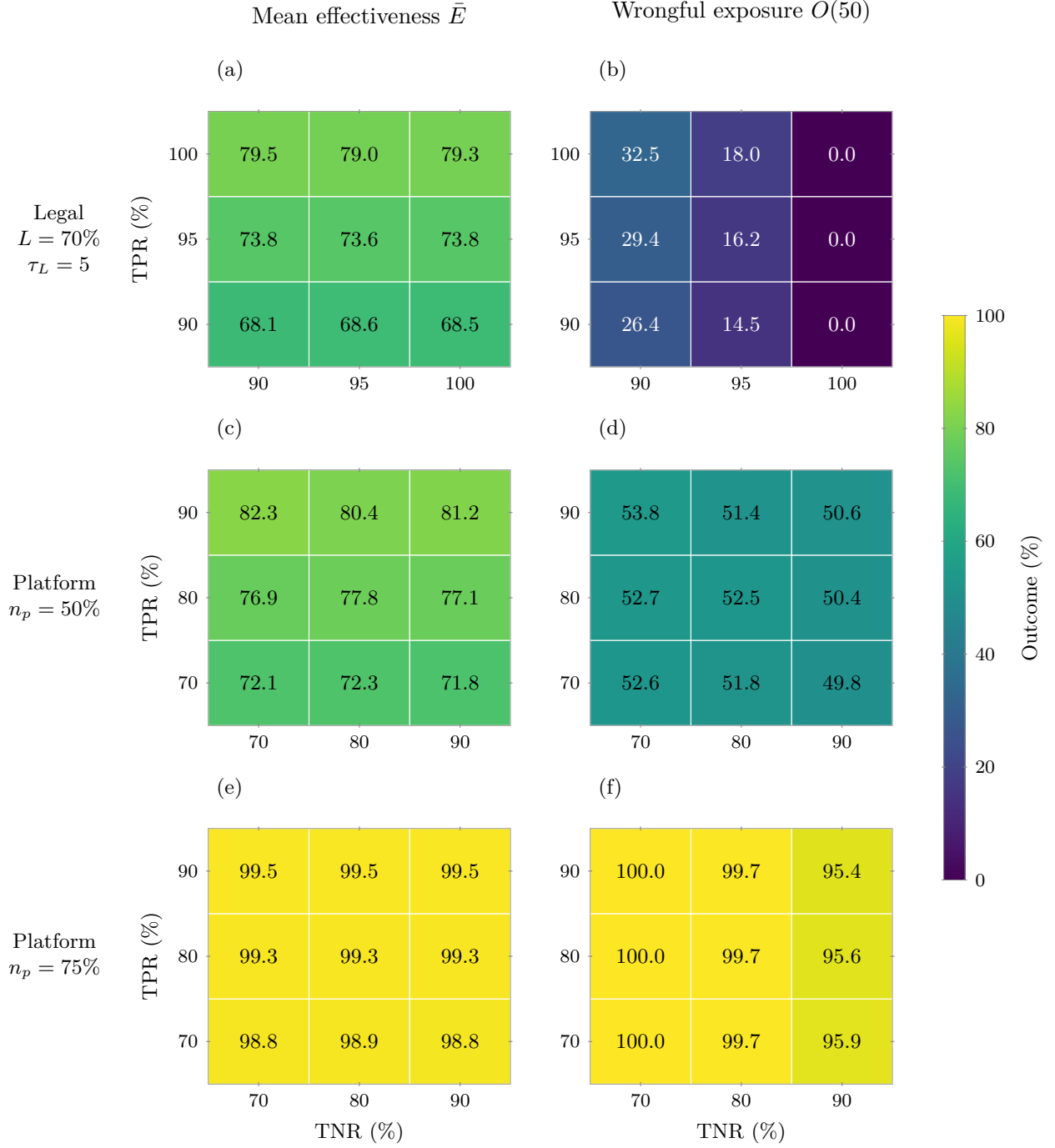


Figure S3: Independent TPR and TNR variation at $p_0 = 30\%$. Design, cell summaries and colour scale follow Fig. S2.

$$p_0 = 50\%$$

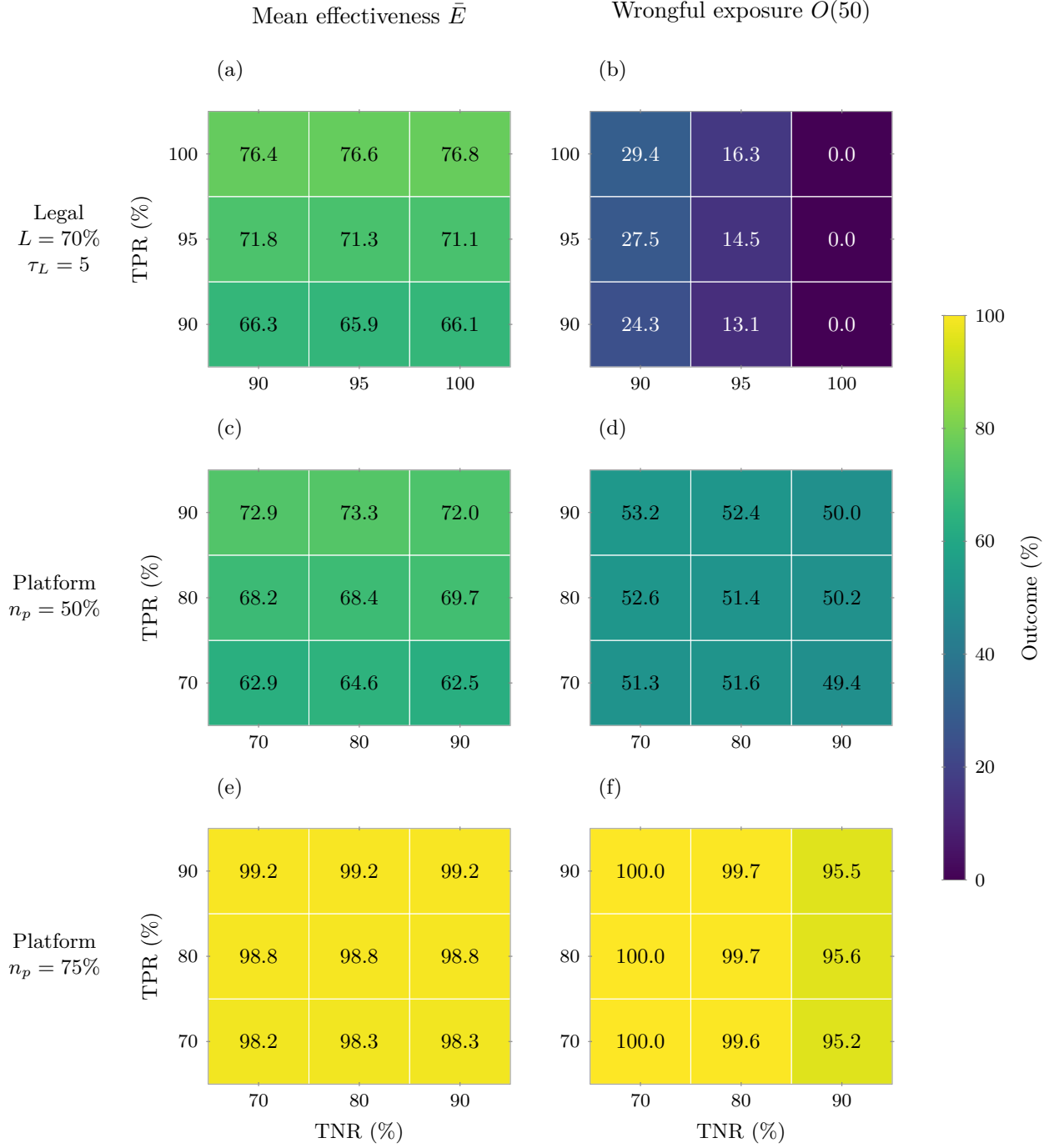


Figure S4: Independent TPR and TNR variation at $p_0 = 50\%$. Design, cell summaries and colour scale follow Fig. S2.

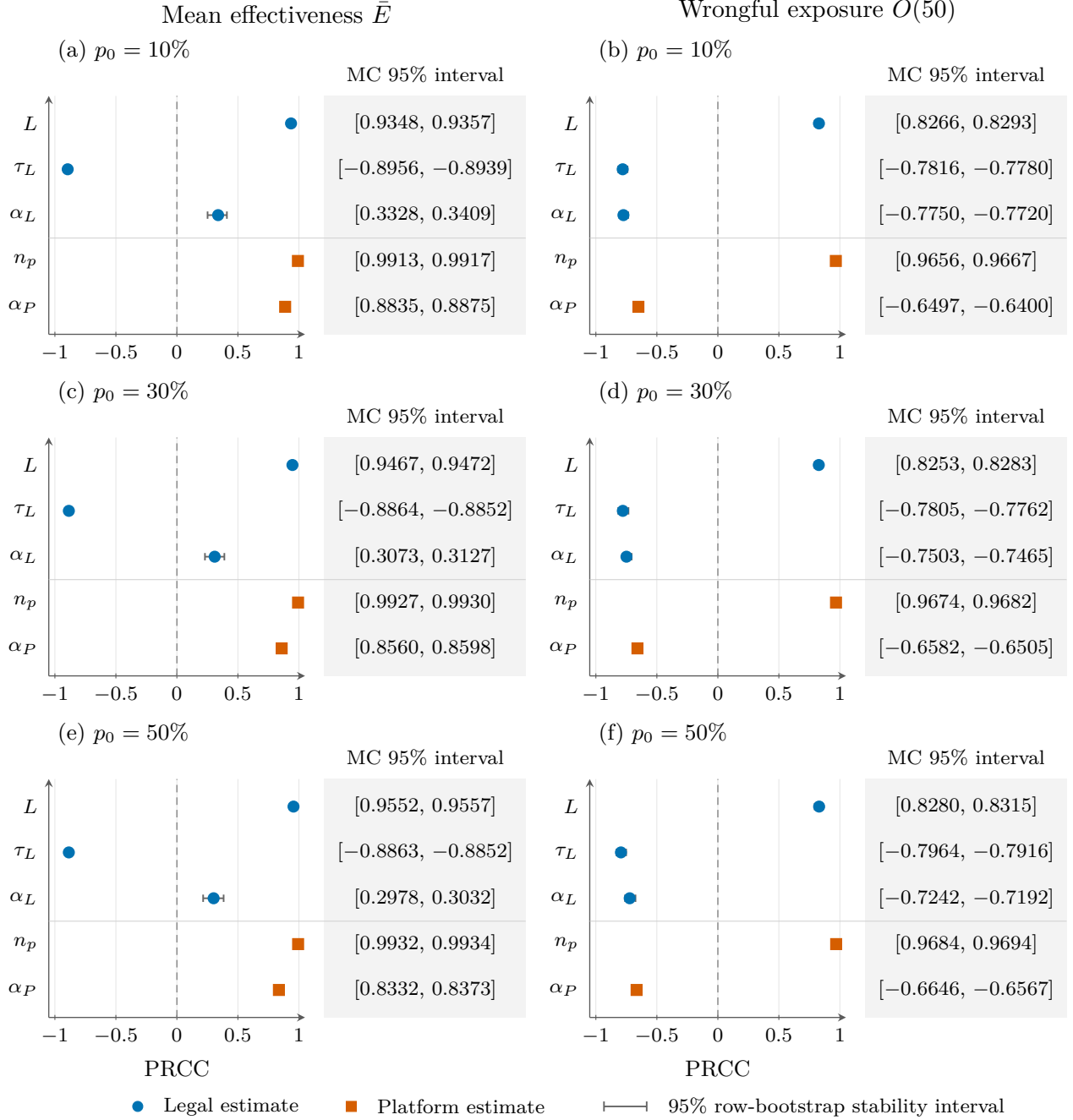


Figure S5: Partial rank correlations (PRCC). Points and whiskers show estimates and marginal 95% row-bootstrap stability intervals; shaded columns report separate 95% within-point Monte Carlo intervals. Each bootstrap uses 1,000 resamples. Row resampling does not preserve LHS strata. PRCC describes conditional rank association; directional interpretation relies on monotonicity and does not replace the primary PAWN analysis.

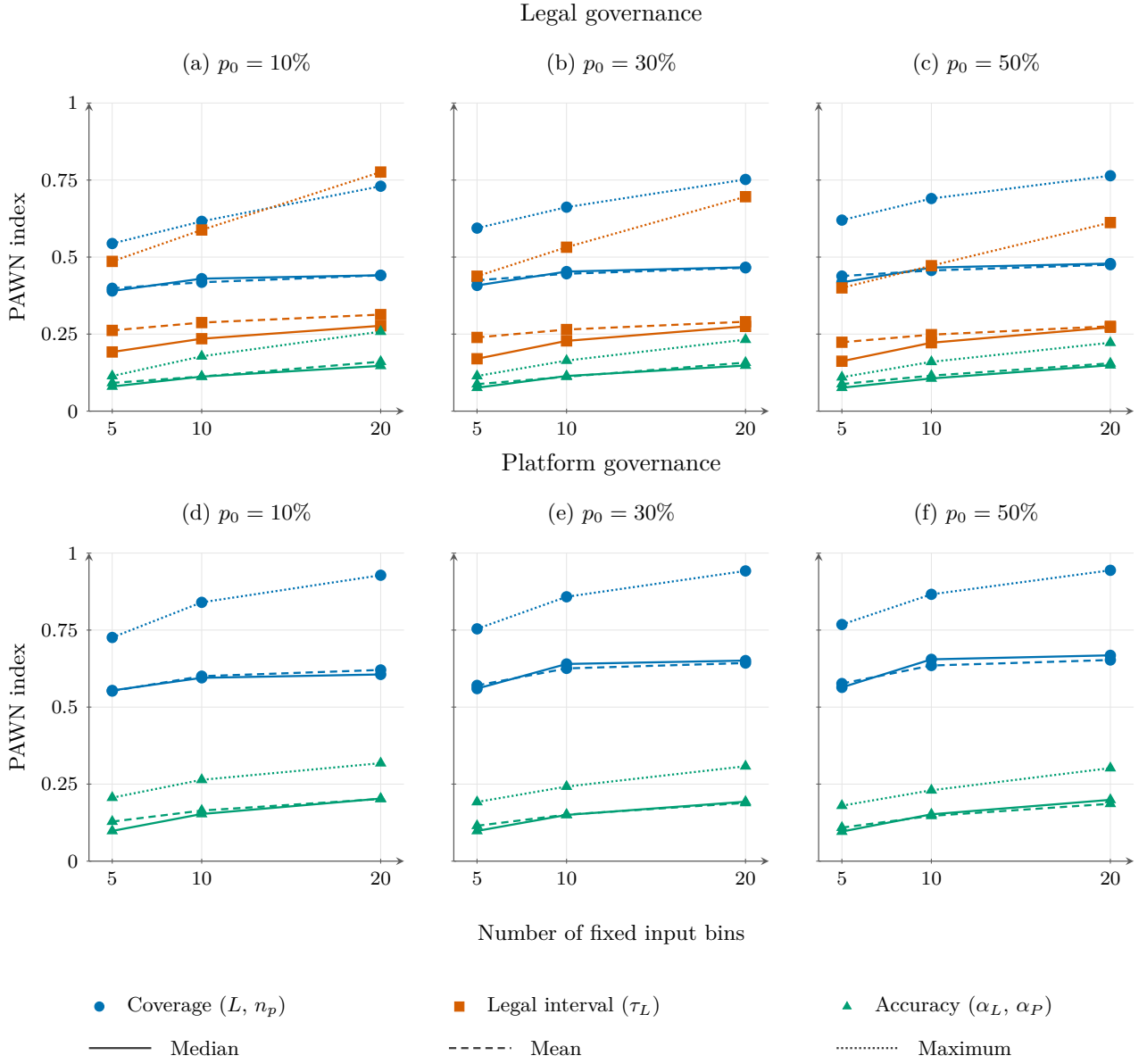


Figure S6: PAWN diagnostics for mean effectiveness $\bar{E}$. Point estimates compare 5, 10 and 20 fixed input bins with median, mean and maximum aggregation; the primary estimator remains the 10-bin median. Full bootstrap intervals, 20-versus 30-run estimates and rankings are provided in `sensitivity_diagnostics.csv`.

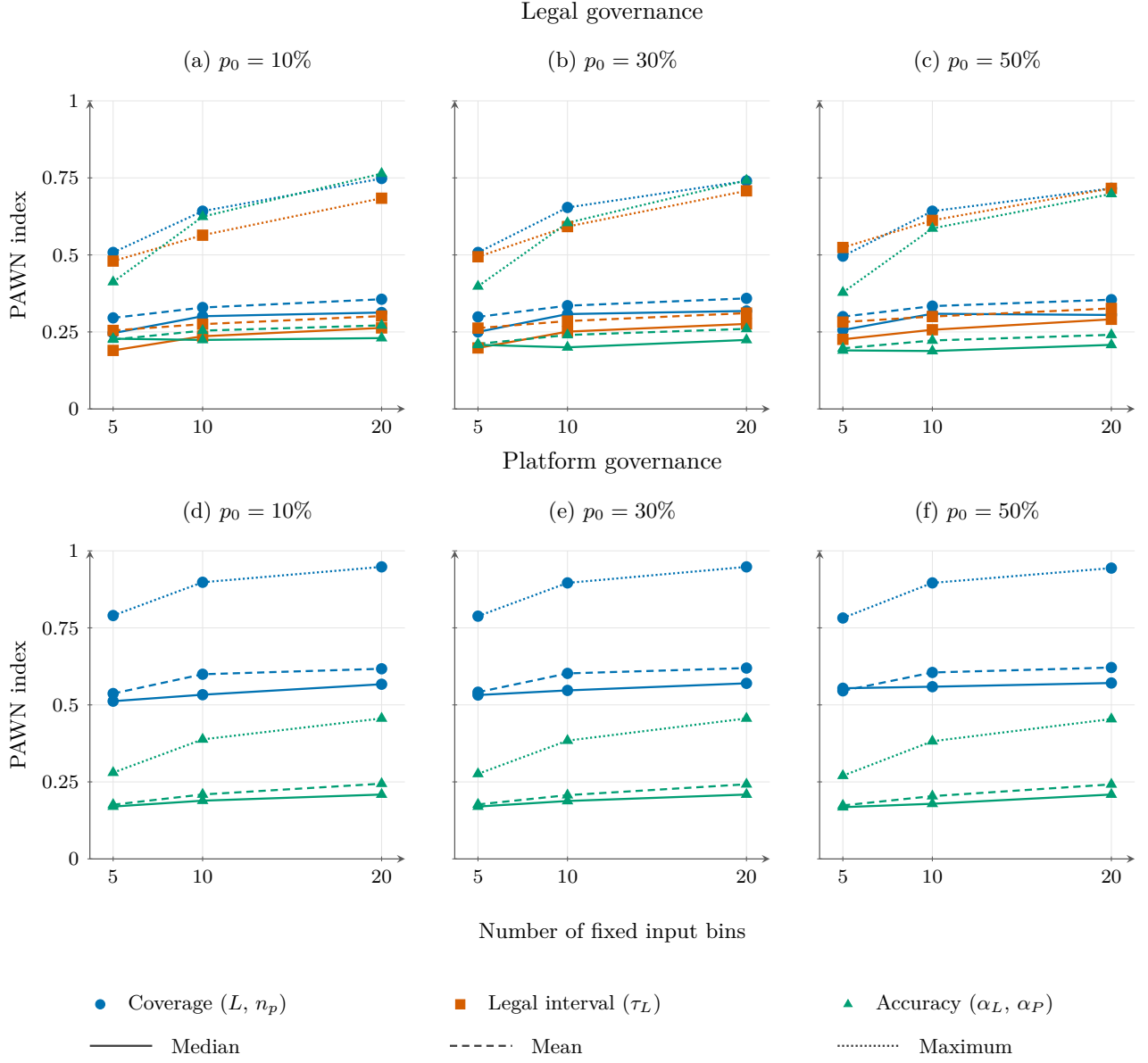


Figure S7: PAWN diagnostics for wrongful exposure $O(50)$. Settings and graphical encodings follow Fig. S6.

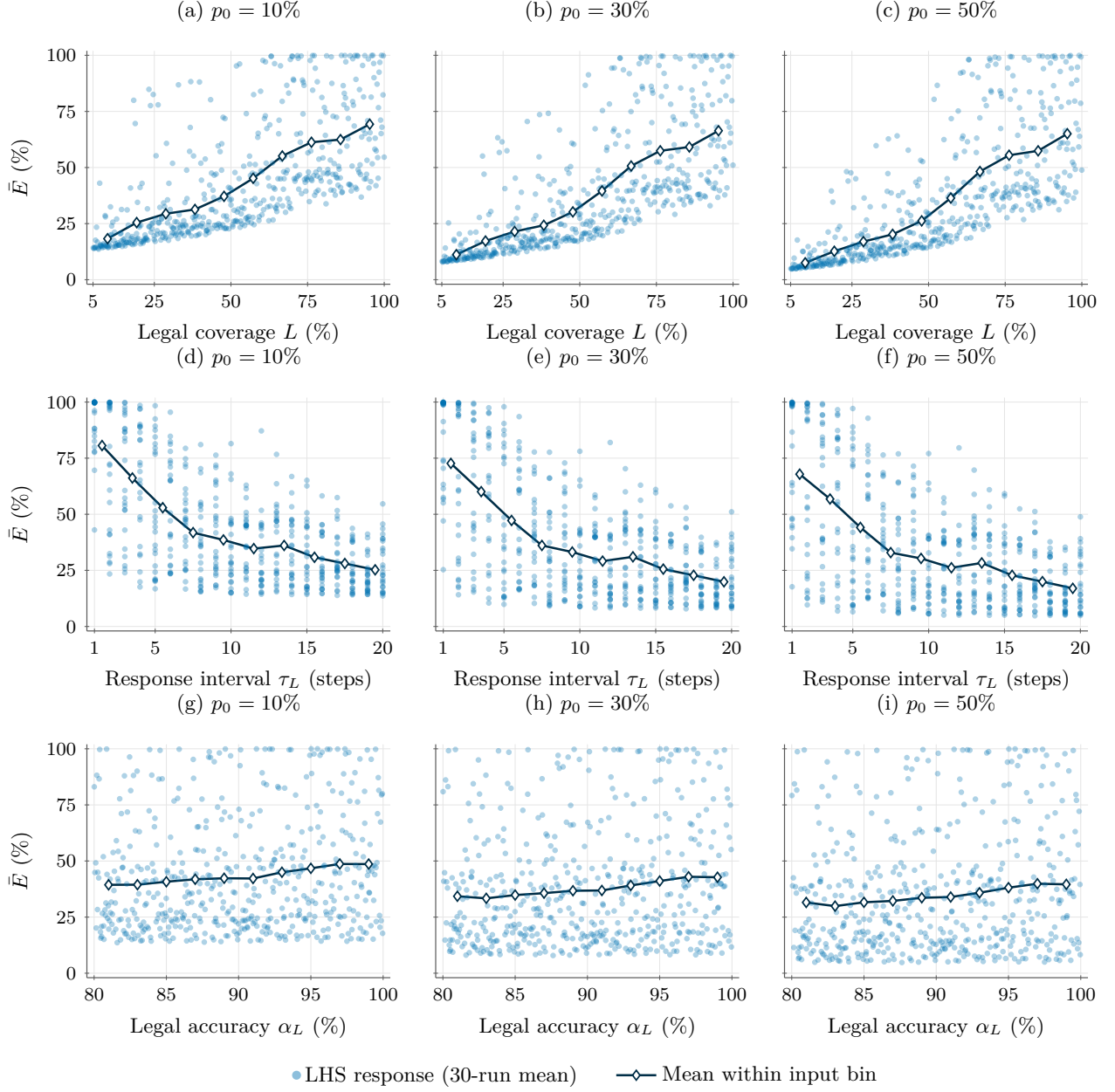


Figure S8: Input–response patterns for legal governance and mean effectiveness $\bar{E}$. Each panel shows 500 LHS responses, each averaged over 30 runs; other inputs vary simultaneously. Lines connect means within 10 fixed input bins. These descriptive summaries are neither isolated single-input causal curves nor confidence bands; they help assess saturation, discreteness and monotonicity.

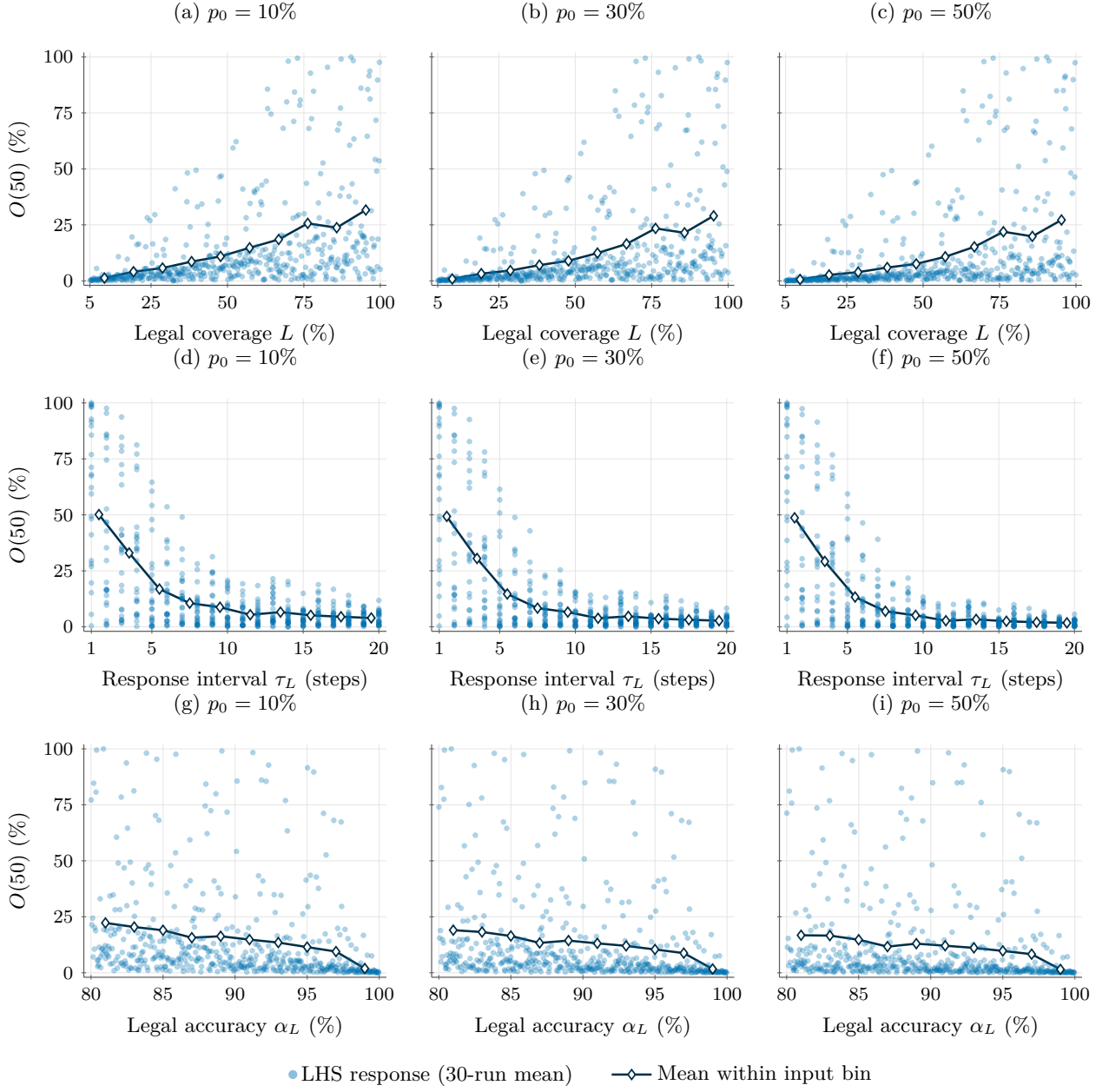


Figure S9: Input-response patterns for legal governance and wrongful exposure $O(50)$. Sampling, bin summaries and interpretation follow Fig. S8.

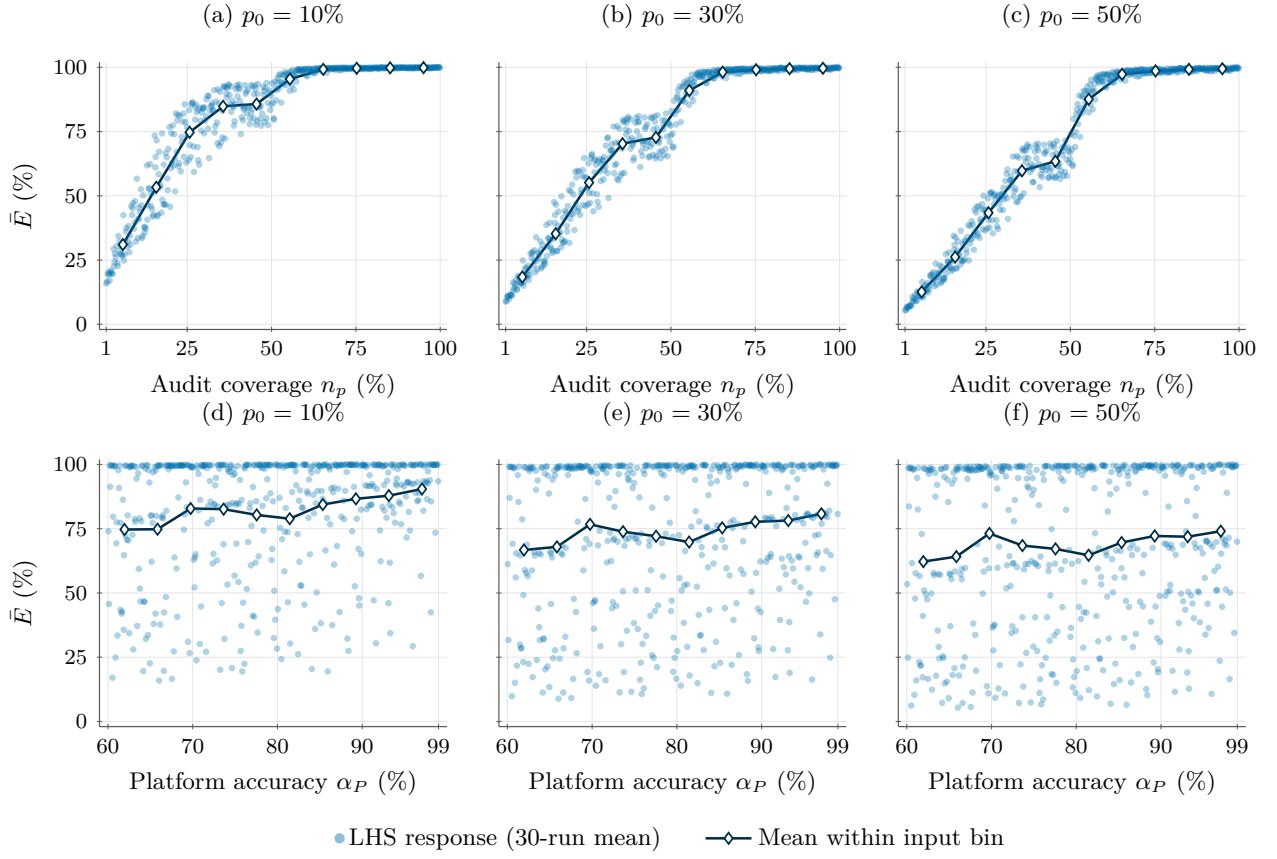


Figure S10: Input–response patterns for platform governance and mean effectiveness $\bar{E}$. Sampling, bin summaries and interpretation follow Fig. S8.

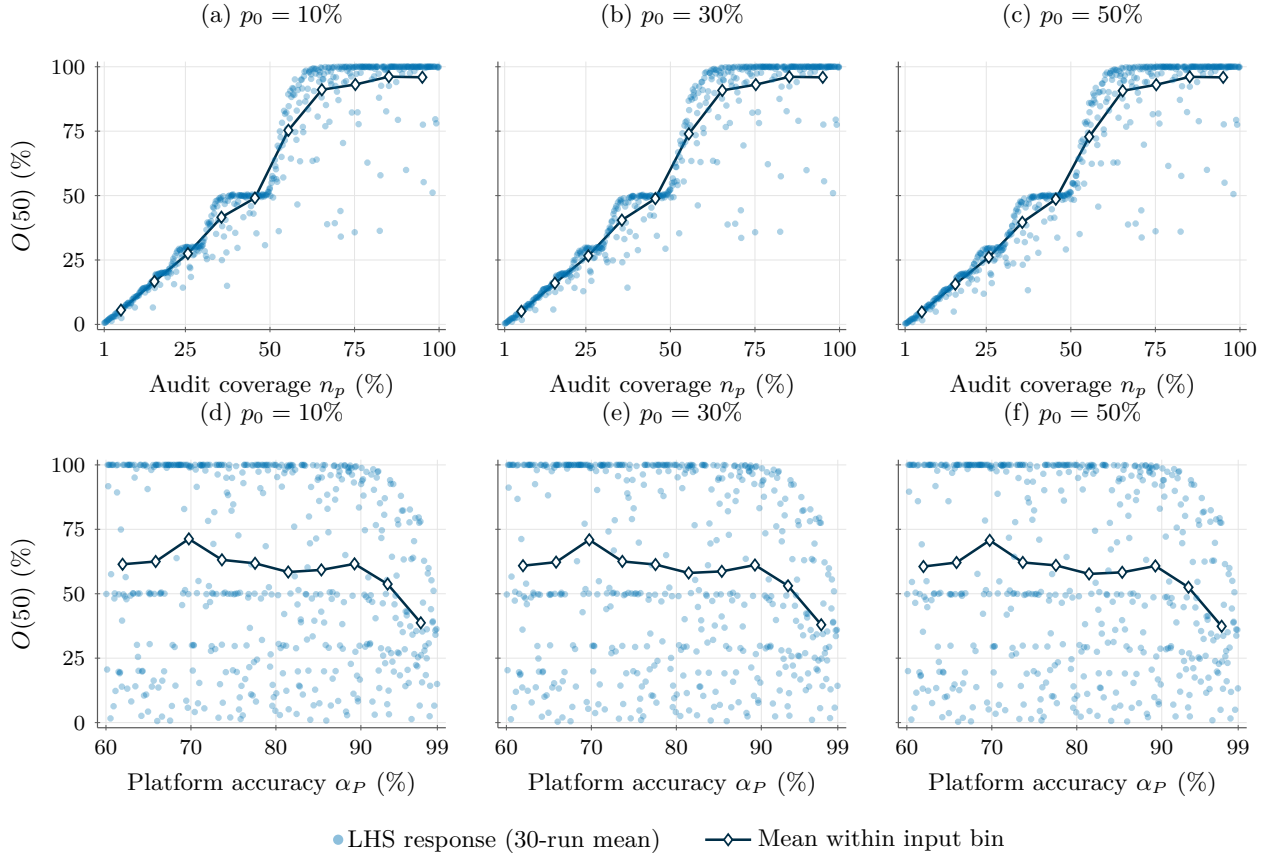


Figure S11: Input–response patterns for platform governance and wrongful exposure $O(50)$. Sampling, bin summaries and interpretation follow Fig. S8.